

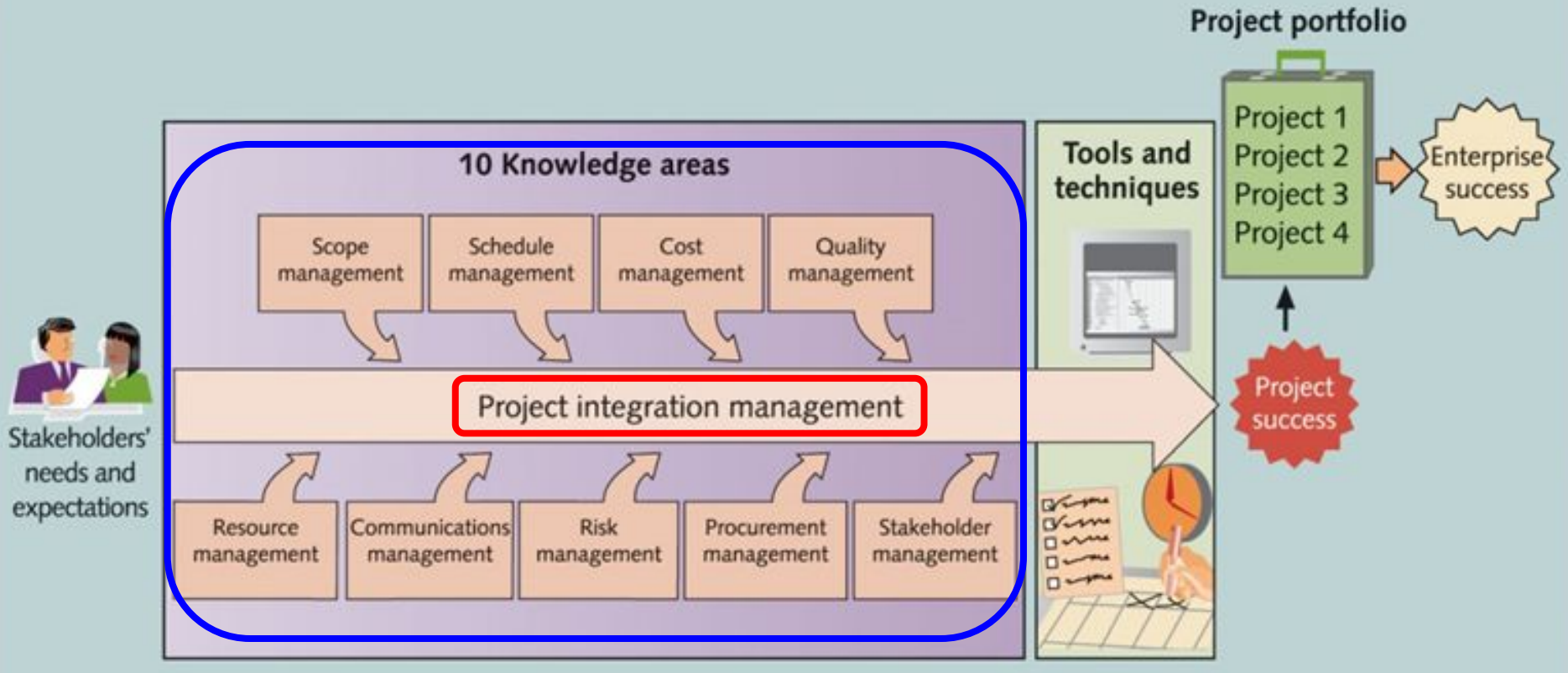
IT Project Management

Lecture 9: Integrating All Knowledge Areas

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All Knowledge Areas



About this lecture

-  Learning objectives:
- Explain the purpose of each remaining PMBOK knowledge area: *resource, communication, procurement, stakeholder, integration*
 - Understand how these areas interact with the core knowledge areas: *scope, time, cost, quality, risk*
 - Apply integration thinking to manage complex projects more effectively
 - Identify interdependencies and avoid isolated project planning

Integrating All Knowledge Areas

Not Just the Triple Constraint

- Beyond the big three **Scope, Time, Cost**
- Projects may also fail due to:
 - Poor communication
 - Resource conflicts
 - Vendor issues
 - Stakeholder resistance
 - etc.

→ *Integration ensures all areas work together*

What Does Integration Mean?

Integration = Coordinating all parts of a project to:

- Align with objectives
- Resolve conflicts between areas
- Manage dependencies
- Handle changes effectively

Example:

Scope change requires updates to *schedule, cost, resources, and communication plan, etc.*

Integration: Intuitive Example

Think of a project like an orchestra:

- Scope = melody
- Schedule = tempo
- Cost = budget
- HR = musicians
- Communications = conductor's cues

→ Integration = music conducting to keep the whole orchestra in sync and produce a beautiful symphony

Resource Management

What is Resource Management?

- Ensures the **right resources** are available and productive
- Balances workloads and resolves conflicts

Two main types of resources:

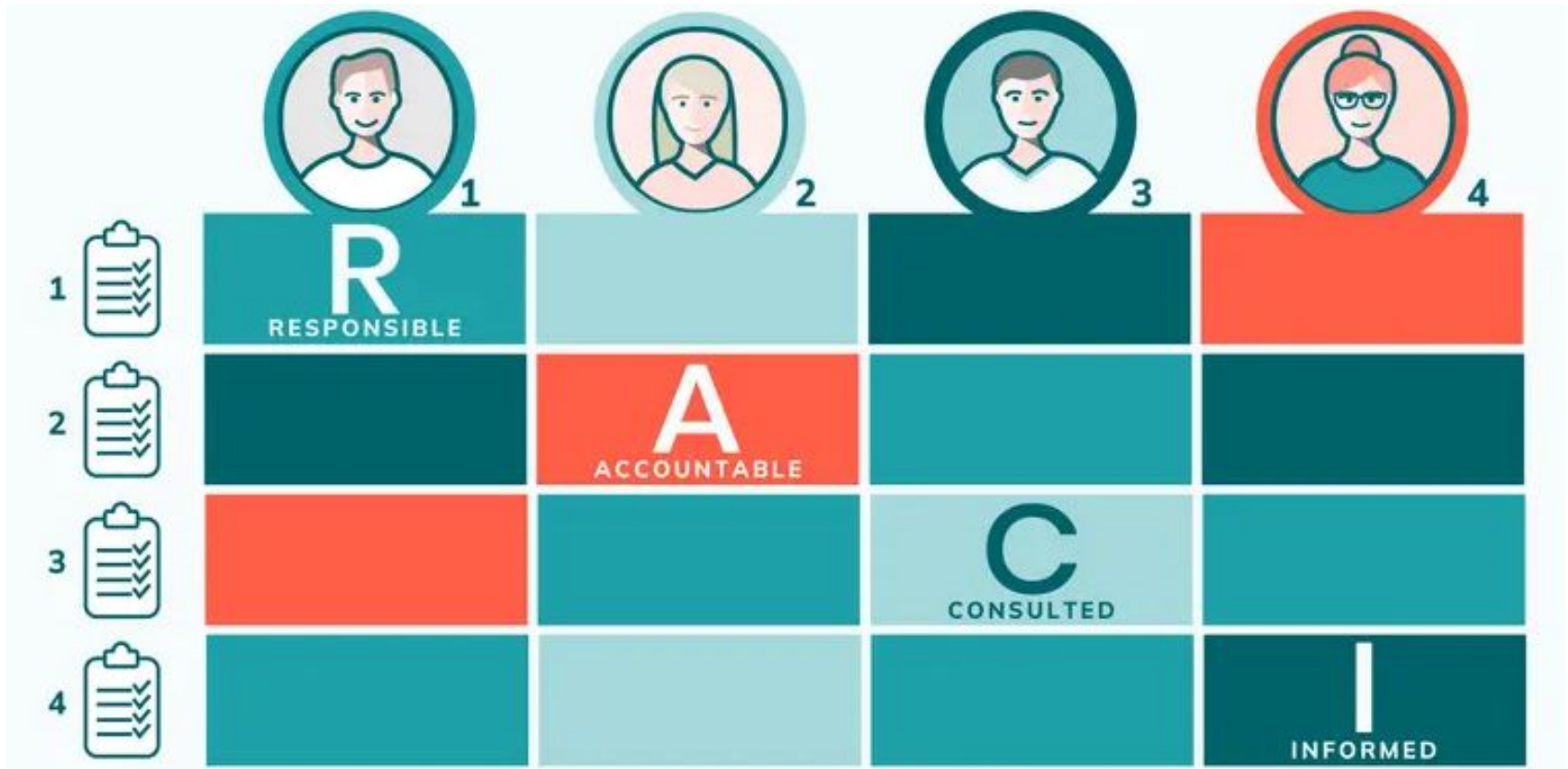
- **Human** (team members, skills, experts)
- **Physical** (equipment, tools, materials)

Resource Planning Tools

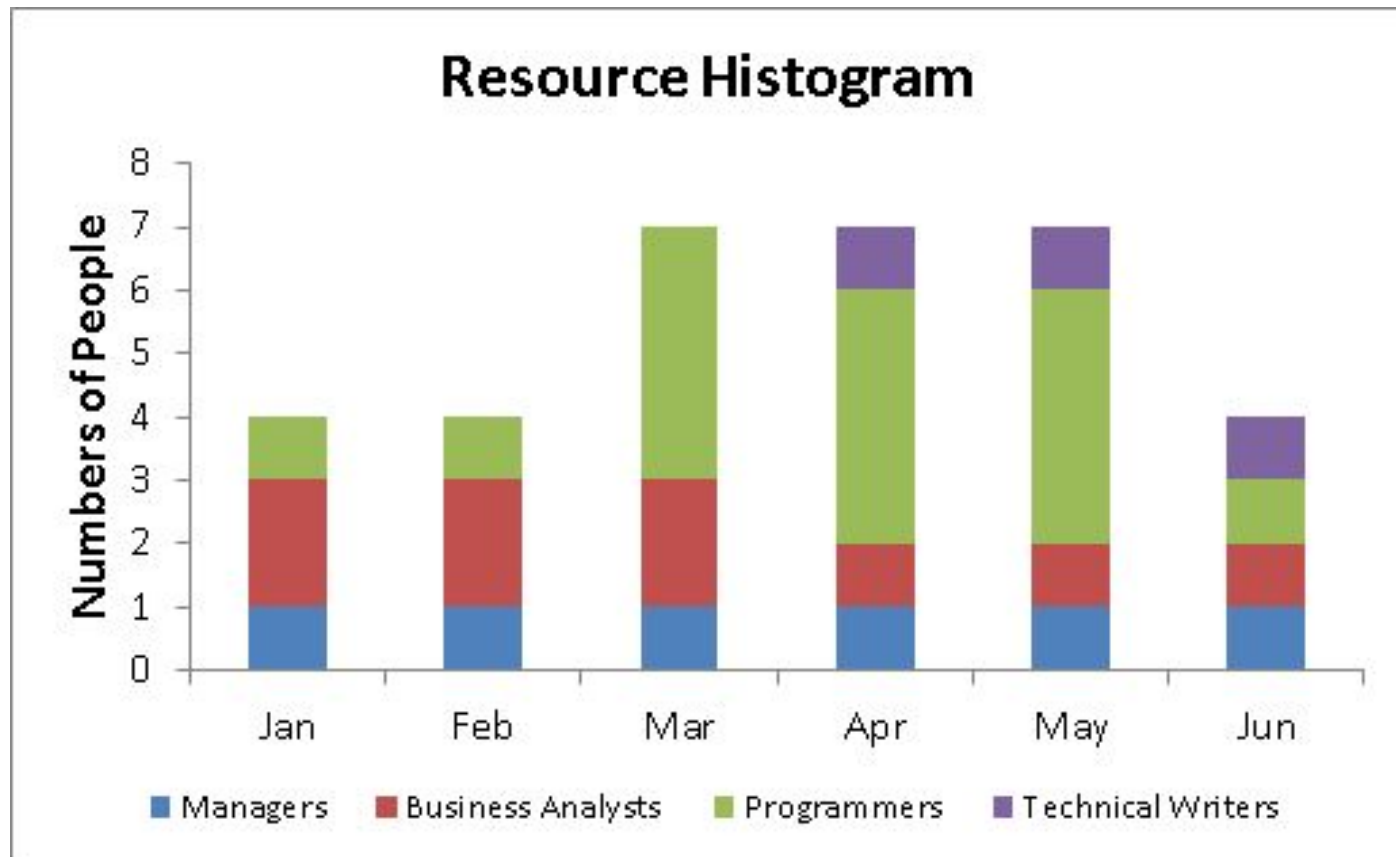
Key Tools:

- **RACI Matrix**
(Responsible, Accountable, Consulted, Informed)
- **Resource Histograms**
(Graph showing how much resources are available)
- **Workload Charts**
(Daily/weekly task assignments by person or team)

Resource Planning Tools: **RACI Matrix**



Resource Planning Tools: Resource Histograms



Allocate Resources:

Tuckman's Model of Team Development

Tuckman's Model of Team Development: 5 Stages

1. **Forming**: Team meets
2. **Storming**: Conflicts arise
3. **Norming**: Start working together
4. **Performing**: Efficient collaboration
5. **Adjourning**: Project ends, team dissolves

Conflict Resolution Techniques

Conflicts happen, due to workload, roles, personality clashes

→ *May ruin the team and the project!*

Common Resolution Methods:

- **Confronting (Problem Solving):** Best, but time-consuming
- **Compromising:** Partial satisfaction
- **Smoothing:** Temporarily minimizes
- **Forcing:** Quick, but risky
- **Ignoring:** Not recommended unless issue is trivial

Communication Management

What is Communication Management?

Ensures project info is **planned, delivered, and understood**

- Crucial for stakeholder engagement and team alignment
- Poor communication is a **top cause of project failure**

→ *Right Message. Right Person. Right Time.*

Communication Planning

Components of a Communication Plan:

- **Stakeholder communication needs**
- **Information type & frequency**
- **Methods/channels** (email, meetings, dashboards)
- **Responsibility** for sending/receiving info

→ Who Needs What, When, and How?

*Example **Communication Matrix**:*

Stakeholder	Info Needed	Frequency	Channel	Provider
Client Sponsor	Status updates	Weekly	Email	PM
Development Team	Task assignments	Daily	Scrum	Scrum Master

Information Flow & Noise

- **Noise** = anything that distorts communication
 - E.g.: language barriers, unclear instructions, tech issues
- **Solution:**
 - **Feedback loops** to ensure clarity: **ask, confirm, and adjust**

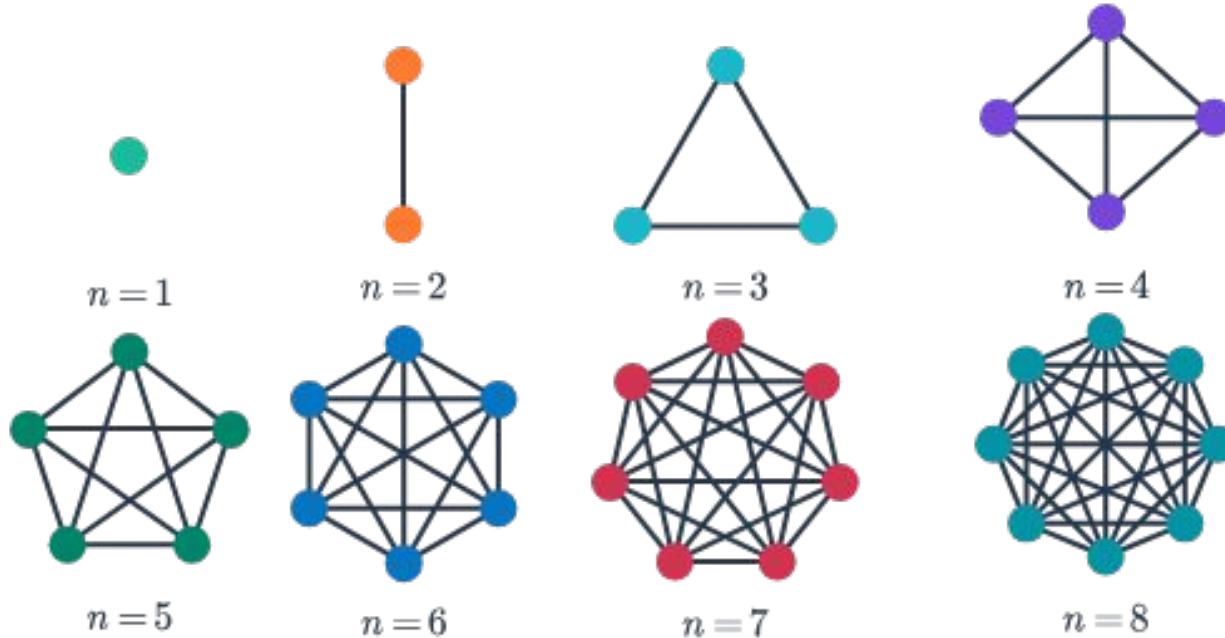
Tips:

- Use visuals and summaries
- Adjust style for audience (execs vs. technical team)
- Confirm understanding (feedback is key)

Problems of Communication Channels

The More People, the More Complex:

Number of communication channels = $n(n-1)/2$



Procurement Management

What is Procurement Management?

Procurement involves **obtaining goods, services, or resources** from outside

- Key to managing **cost, time, and quality** when outsourcing
- Common in IT, construction, manufacturing projects

→ *Working with external partners*

Example:

Hiring a software vendor, renting equipment, or purchasing cloud services

Make-or-Buy Decision

Factor	Make (In-House)	Buy (Outsource)
Cost efficiency	Bad	Good
Strategic Control	Good	Bad
Speed	Bad	Good
Expertise required	Bad	Good

Common contract types

Type	Description	Risk to Buyer	Risk to Seller
Fixed-Price (Lump Sum)	Pre-agreed cost regardless of actual effort	Low risk low return	High risk high return
Time & Materials (T&M)	Pay per hour or item used	Medium	Medium
Cost-Reimbursable	Reimbursed for actual costs	High risk high return	Low risk low return

Vendor Evaluation & Selection

Factors to consider:

- Cost proposal
- Experience & references
- Technical capabilities
- Delivery timelines
- Legal/contractual terms

→ Use **Weighted Scoring Models** to evaluate vendor bids

Stakeholder Management

What is Stakeholder Management?

Reminder: Stakeholders are anyone impacted by the project

E.g., customers, sponsors, team members, users, suppliers, regulators, etc.

Effective stakeholder management ensures to:

- Identify, understand, and manage expectations
 - Get relevant people involved
 - Reduced resistance and conflicts

Stakeholder Analysis

- **Interest:** *How much they care about project outcomes?*
- **Influence:** *How much power they have over the project?*

Example:

- **Project Sponsor** = High power, high interest → *Manage Closely*
- **Government Agency** = High power, low interest → *Keep Satisfied*

	Low Power	High Power
High Interest	Keep Informed	Manage Closely
Low Interest	Just Monitoring	Keep Satisfied

Stakeholder Engagement Strategies

Choose tactics based on their quadrant:

- **Manage Closely**
 - → Frequent meetings, detailed reports
- **Keep Satisfied**
 - → High-level updates, consultation when needed
- **Keep Informed**
 - → Newsletters, progress emails
- **Monitor**
 - → Occasional check-ins

	Low Power	High Power
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Communication & Expectation Alignment

Tools for aligning stakeholders' expectations:

- **Active listening**
- **Clear scope definitions**
- **Early involvement**
- **Feedback loops**

Warning signs to be identified early:

- Conflicting goals
- Mismatched timelines
- Last-minute objections

→ Use feedback and negotiation to realign early

Integration Management

What is Integration Management?

Project Integration Management ensures:

- All knowledge areas work **together seamlessly**
- A **unified direction** for the project
- Decisions are made **holistically**, not in isolation

→ *The glue that holds it all together*

Without integration:

- Scope, time, and cost plans **conflict**
- Changes create **chaos**
- Lessons learned are **lost**

Integration Management In Project Starting

Project Charter:

- Authorizes the project
- Names project manager
- Summarizes objectives, stakeholders, budget

Project Management Plan:

- Detailed roadmap
- Combines scope, schedule, cost, quality, resources, risks, etc.

⇒ Many aspects are integrated

Integration Management in Change Control

Change control's key elements:

- **Change Request** → Logged and evaluated
- **Impact Analysis** → On scope, time, cost, quality
- **Change Control Board (CCB)** → Approves or rejects
- **Update Plan** → Communicate to all stakeholders

⇒ Many aspects are integrated

Integration Management in Project Closing

Project closure ensures:

- All deliverables are accepted
- Contracts are finalized
- Documents and lessons learned are archived
 - Transfer knowledge for future projects
 - Avoid “reinventing the wheel”

⇒ Full project cycle, from project starting to closing and back to starting

Putting It All Together

The Big Picture

Project success depends on **coordination** across all 10 knowledge areas.

For example:

- You can't manage **cost** without considering **resources**
- You can't manage **scope** without communicating it to **stakeholders**

→ *No area can stand alone*

Systems Thinking

Projects as living systems:

- Every decision has **ripple effects**
- Must think in **loops** not just **linear plans**
- Integration is not just managing tasks: it's managing **interdependencies**

Example:

Changing a feature in an app → affects UI, testing, timeline, costs, vendor coordination, and customer expectations

Project Management Plan (PMP)

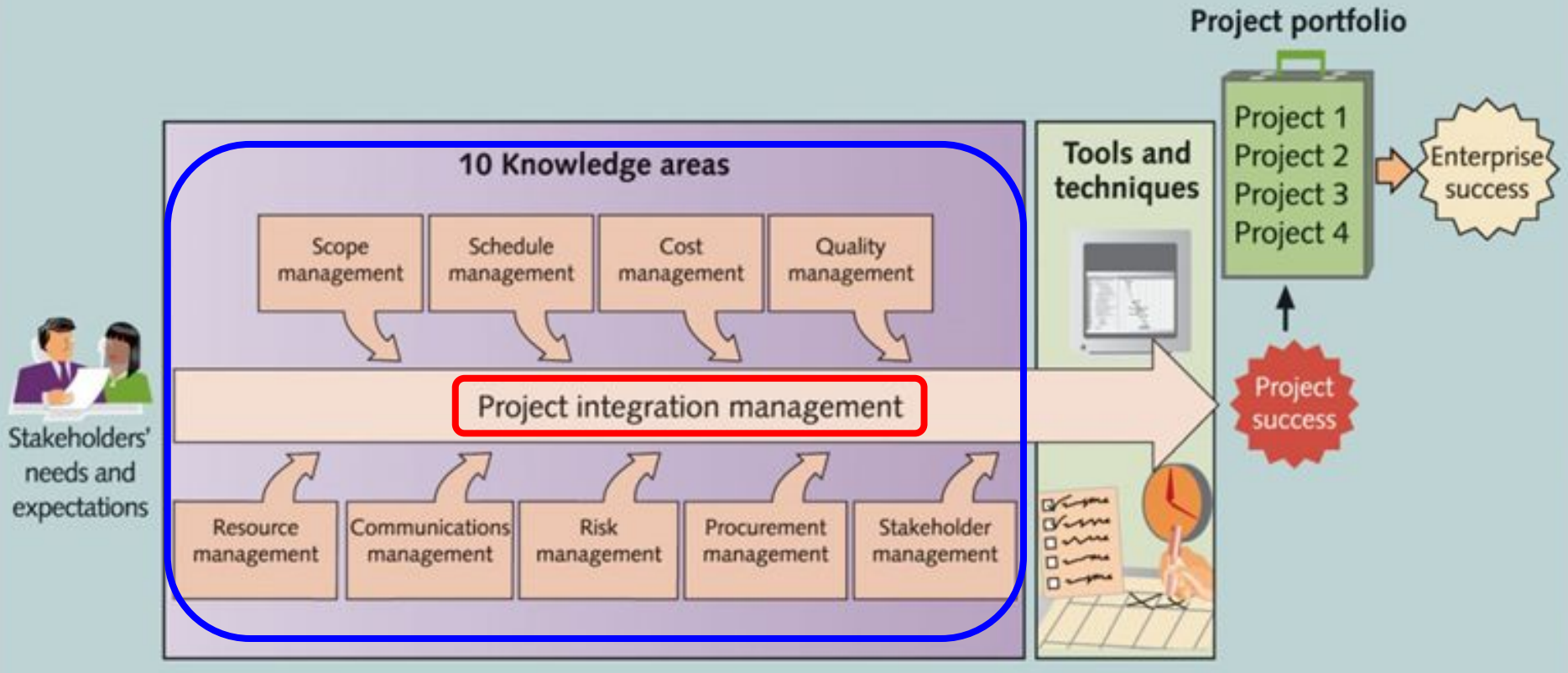
The PMP includes:

- *Scope baseline*
- *Schedule baseline*
- *Cost estimates*
- *Risk strategies*
- *Quality profiles*
- Communication & stakeholder plans
- Procurement strategies
- Integration framework

→ Ensures all knowledge areas are aligned

→ **Not static**: must be updated as the project evolves

All Knowledge Areas



Exercise

Exercise 1

Scenario: *You're launching a new ride-sharing app, like Uber or Grab.*

→ *Assign responsibilities (who) and identify integration points (what they need to do) across all knowledge areas.*

Knowledge Area	Key Question
Integration Management	Who owns the overall coordination and change control?
Scope Management	What exactly does the app need to do? Who defines the features?
Schedule Management	How will we manage sprints and deadlines?
Cost Management	How will the project be funded? Who tracks expenses?
Quality Management	How do we ensure the app works as intended and meets customer expectations?
Resource Management	Who's on the team? Do we have the right skills and tools?
Communication Management	How will stakeholders (investors, users, developers) stay informed?
Risk Management	What could go wrong? How do we prepare?
Procurement Management	Will we outsource any services (e.g., cloud, design)? How do we choose vendors?
Stakeholder Management	Who are the key stakeholders, and how do we keep them engaged?

Answer

Knowledge Area	Responsibility / Key Integration Point
Integration Management	Project Manager (PM) – Oversees coordination, manages change control board, ensures plan alignment.
Scope Management	Product Owner + Business Analyst – Defines app features (e.g. ride-matching, payment, rating system).
Schedule Management	Scrum Master / PM – Develops sprint timelines, sets milestones (beta launch, QA phase, final release).
Cost Management	Finance Lead / PM – Budgets for development, marketing, cloud services; tracks burn rate and variances.
Quality Management	QA Lead – Implements test plans, usability testing, automated bug checks, ensures performance standards.
Resource Management	PM + HR – Assigns front-end/backend/mobile devs, hires UX/UI freelancers, balances workload.
Communication Management	PM + Comms Lead – Manages updates to devs, investors, and users; maintains Slack, emails, status reports.
Risk Management	PM + Risk Analyst – Identifies risks (e.g. API downtime, data breaches, driver shortages), sets responses.
Procurement Management	Operations/Procurement Officer – Selects cloud provider (e.g. AWS), outsources map services or dev ops.
Stakeholder Management	PM + Stakeholder Liaison – Engages riders, drivers, regulators, investors through surveys and demos.

Summary

Summary

-  You have learned to:
- Explain the purpose of each remaining PMBOK knowledge area: *resource, communication, procurement, stakeholder, integration*
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Thank you for listening

Q & A